E-Commerce Sales and Customer Behavior Analysis

Project Overview

I analyzed an e-commerce relational database using MySQL Workbench. The database consisted of six tables covering customers, products, orders, order items, payments, and refunds.

The objective of this project was to validate the dataset, examine relationships between tables, and analyze sales performance, product performance, customer spending, payment methods, and refunds to identify meaningful business insights.

Tools and Techniques

Database & Analysis

MySQL Workbench

Data Validation

Row count validation

Missing-value (NULL) checks

Duplicate ID checks

Table relationship validation

Data type and structure inspection

SQL Techniques

INNER JOIN

LEFT JOIN

SUM()

COUNT()

COUNT(DISTINCT)

GROUP BY

ORDER BY

LIMIT

DATE()

DATE_FORMAT()

Subqueries

Key Findings and Business Recommendations

1. Net Product Revenue and Order Volume

Finding:

The analysis reported $54,033,377.13 in net product revenue across 89,114 completed orders. Revenue was calculated by summing the line_total column from the order_items table. The line_total values account for item quantities and discounts. Shipping fees were excluded to focus specifically on net product sales.

Business Recommendation:

Monitor revenue and order volume over time. Additional cost and expense data would be required to evaluate overall profitability.

2. Monthly Revenue Trends

Finding:

Monthly revenue remained relatively stable throughout 2023, ranging from approximately $1.3 million to $1.6 million.
Business Recommendation:
Monthly revenue trends can support sales forecasting and inventory planning. Additional years of data would provide a stronger basis for identifying long-term trends and seasonality.

3. Product Revenue Distribution
Finding:
Apparel Item 0733 was the highest-ranked product in the analysis, generating $125,822.57, representing approximately 0.23% of reported net product revenue.
This indicates that revenue was distributed across a wide range of products rather than being heavily dependent on one product.
Business Recommendation:
Compare product revenue, sales volume, and profit margins to identify high-performing products and opportunities for improvement.

4. High-Spending Customers
Finding:
The top 10 customers by spending were predominantly based in the United States. The highest-spending customer generated $12,811.69 across 14 orders.
Business Recommendation:
Investigate repeat-purchase behavior and consider targeted retention initiatives, such as loyalty programs, for high-value customers.

5. Refund Analysis
Finding:
The database contained 79,110 refund records with a reported total refund amount of $3,387,902.65. This represents approximately 6.27% of reported net product revenue.
Business Recommendation:
Validate refund statuses and record definitions before calculating a final refund rate. Once validated, refund reasons and patterns can be investigated to identify opportunities to reduce avoidable refunds.

6. Payment Method Data Quality
Finding:
Card was the dominant recorded payment method, accounting for approximately $38.3 million. However, the value "Completed" also appeared in the payment_method column, indicating inconsistent or potentially misclassified data.
Business Recommendation:
Investigate the source of the inconsistent values, standardize payment-method categories, and rerun the payment analysis using validated records.
Data Validation
Before conducting the analysis, several data validation checks were performed:
Verified the number of records in each table

Checked critical fields for missing values
Checked for duplicate order IDs
Examined relationships between related tables
Inspected table structures and column definitions
Verified the calculation of line_total
These checks helped establish that the dataset was suitable for the subsequent analysis while also identifying potential data-quality issues.
Repository Contents

The repository contains:
Ecommerce_Sales_Analysis.sql
Contains the SQL queries used for database verification, data validation, relationship analysis, sales analysis, customer analysis, product analysis, payment analysis, and refund analysis.
Screenshots/
Contains screenshots showing the database structure, validation checks, SQL queries/results, and revenue calculation verification.

Conclusion
This project demonstrates how SQL can be used to inspect and validate a relational database, analyze sales and customer behavior, evaluate product performance, and identify data-quality issues.
The analysis also demonstrates the importance of validating business definitions and data inconsistencies before using analytical results to make business recommendations.